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Part 1:

Q1.

Docker Containers Running - Command: docker ps

- Worker (consumer) service started (shows: "Consumer is ready and listening...")
- FastAPI service running (shows server started on http://127.0.0.1:8000)

kafka installed

```
(ds) Abhinata Sanabada ~ % cd /Users/tachyon_hunter/Desktop/sem-3/DS-Slides/DS/Homework-submission/Homework-8
(ds) Abhinata Sanabada Homework-8 % docker compose up -d
WARNING: /Users/tachyon_hunter/Desktop/sem-3/DS-Slides/DS/Homework-submission/Homework-8/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 13/15
  ✓ zookeeper Pulled
    ✓ b4d293195cb8 Pull complete
    ✓ e7137cfa206e Extracting 1 s
    ✓ 9855f0939acd Pull complete
    ✓ fddef29ac805 Pull complete
    ✓ 1323a34717bc Extracting 10 s
    ✓ 1c920487b3ee Pull complete
    ✓ 40e837380f80 Pull complete
    ✓ 0feac35068e9 Pull complete
    ✓ 2ff788fc6a64 Pull complete
    ✓ 4b3ca0422b2d Pull complete
    ✓ 04d70d2a3c8c Pull complete
  ✓ kafka Pulled
    ✓ 9c2f80a84992 Pull complete
    ✓ e25648cb50ed Pull complete
[+] Running 3/3
  ✓ Network homework-8_default Created
  ✓ Container homework-8-zookeeper-1 Started
  ✓ Container homework-8-kafka-1 Started
(ds) Abhinata Sanabada Homework-8 % docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                                NAMES
f6664b32b5a6   confluentinc/cp-kafka:7.5.0        "/etc/confluent/dock_  48 seconds ago Up 47 seconds 0.0.0.0:9092->9092/tcp, [::]:9092->9092/tcp homework-8-kafka-1
6584b36698d4   confluentinc/cp-zookeeper:7.5.0    "/etc/confluent/dock_  49 seconds ago Up 47 seconds 0.0.0.0:2181->2181/tcp, [::]:2181->2181/tcp homework-8-zookeeper-1
(ds) Abhinata Sanabada Homework-8 %
```

Kafka created and run

☒ Only show running containers

☐	Name	Container ID	Image	Port(s)	Actions
☐	homework-8	-	-	-	↕ 🔴 ⋮ 🗑
☐	zookeeper-1	6584b36698d4	confluentinc/cp-zookeeper:7.5.0	2181:2181	↕ 🔴 ⋮ 🗑
☐	kafka-1	f6664b32b5a6	confluentinc/cp-kafka:7.5.0	9092:9092	↕ 🔴 ⋮ 🗑

User

```
WARNING: Due to limitations in metric names, topics with a period (".") or underscore ("_") could collide. To avoid issues it is best
Created topic user_responses.
(ds) Abhinata Sanabada Homework-8 % docker exec -it homework-8-kafka-1 kafka-topics --list --bootstrap-server localhost:9092
user_requests
user_responses
(ds) Abhinata Sanabada Homework-8 %
```

Q2. Execute **CREATE_USER** and **GET_USER** operations (from homework_demo.py or API call)

Consumer

```
○ (ds) (.venv) Abhinita Sanabada Homework-8 % python worker.py
Consumer is ready and listening...
□
```

Get user

```
BASE_URL = "http://127.0.0.1:8000"

def pretty_print(title: str, response: requests.Response) -> dict:
    print(f"\n===== {title} =====")
    print("Status Code:", response.status_code)
    try:
        data = response.json()
        print(json.dumps(data, indent=2))
        return data
    except Exception:
        print(response.text)
        return {}

def main() -> None:
    create_payload = {
        "operation": "CREATE_USER",
        "name": "Alice",
        "email": "alice@test.com",
        "age": 25,
    }

    create_response = requests.post(
        f"{BASE_URL}/users",
        json=create_payload,
        timeout=20,
    )

    create_data = pretty_print("CREATE_USER", create_response)

    if not create_data.get("success"):
```

```

        print("\nCREATE_USER failed, so GET_USER cannot continue.")
        return

    user_id = create_data["userId"]

    get_payload = {
        "operation": "GET_USER",
        "userId": user_id,
    }

    get_response = requests.post(
        f"{BASE_URL}/users",
        json=get_payload,
        timeout=20,
    )

    pretty_print("GET_USER", get_response)

if __name__ == "__main__":
    main()

```

```

● (ds_venv) Abhinita Sanabada Homework-8 % conda activate ds
○ (ds) (ds_venv) Abhinita Sanabada Homework-8 % python3.11 -m uvicorn app.main:app --reload
INFO: Will watch for changes in these directories: ['/Users/tachyon_hunter/Desktop/sem-3/DS-Slides/DS/Homework-submission/Homework-8']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [67523] using WatchFiles
INFO: Started server process [67563]
INFO: Waiting for application startup.
INFO: Application startup complete.

```

Q3. Show logs of:

```

● (ds) (ds_venv) Abhinita Sanabada app % python homework_demo.py
===== CREATE_USER =====
Status Code: 200
{
  "success": true,
  "userId": "30a9d728-a2cd-4ce4-ae98-d22037615796",
  "message": "User created"
}

===== GET_USER =====
Status Code: 200
{
  "success": true,
  "user": {
    "userId": "30a9d728-a2cd-4ce4-ae98-d22037615796",
    "name": "Alice",
    "email": "alice@test.com",
    "age": 25
  }
}

```

```

File ~/Users/tachyon_hunter/Desktop/sem-3/DS-Slides/DS/Homework-submission/Homework-8/app/main.py, line 7, in <module>
    from kafka import KafkaConsumer, KafkaProducer
ModuleNotFoundError: No module named 'kafka'
^CINFO: Stopping reloader process [66180]
• (ds_venv) Abhinita Sanabada Homework-8 % clear
• (ds_venv) Abhinita Sanabada Homework-8 % conda activate ds
○ (ds) (ds_venv) Abhinita Sanabada Homework-8 % python3.11 -m uvicorn app.main:app --reload
INFO: Will watch for changes in these directories: ['/Users/tachyon_hunter/Desktop/sem-3/DS-Slides/DS/Homework-submission/Homework-8']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [67523] using WatchFiles
INFO: Started server process [67563]
INFO: Waiting for application startup.
INFO: Application startup complete.
[FastAPI] Message sent to user_requests topic: {"correlation_id": "3ede5265-3c97-4b11-b198-2f9991494533", "reply_to": "user_responses", "data": {"operation": "CREATE_USER", "name": "Alice", "email": "alice@test.com", "age": 25}}
[FastAPI] Response received from user_responses topic and returned to client: {"correlation_id": "3ede5265-3c97-4b11-b198-2f9991494533", "data": {"success": true, "userId": "30a9d728-a2cd-4ce4-ae98-d22037615796", "message": "User created"}}
INFO: 127.0.0.1:56004 - "POST /users HTTP/1.1" 200 OK
[FastAPI] Message sent to user_requests topic: {"correlation_id": "5d70ff2a-d0e5-4248-8bb9-510e7a0dd15b", "reply_to": "user_responses", "data": {"operation": "GET_USER", "userId": "30a9d728-a2cd-4ce4-ae98-d22037615796"}}
[FastAPI] Response received from user_responses topic and returned to client: {"correlation_id": "5d70ff2a-d0e5-4248-8bb9-510e7a0dd15b", "data": {"success": true, "user": {"userId": "30a9d728-a2cd-4ce4-ae98-d22037615796", "name": "Alice", "email": "alice@test.com", "age": 25}}}
INFO: 127.0.0.1:56013 - "POST /users HTTP/1.1" 200 OK

```

```

(ds) (.venv) Abhinita Sanabada ~ % cd /Users/tachyon_hunter/Desktop/sem-3/DS-Slides/DS/Homework-submission/Homework-8
(ds) (.venv) Abhinita Sanabada Homework-8 % python worker.py
Consumer is ready and listening...
[Worker] Message received from user_requests: {"correlation_id": "3ede5265-3c97-4b11-b198-2f9991494533", "reply_to": "user_responses", "data": {"operation": "CREATE_USER", "name": "Alice", "email": "alice@test.com", "age": 25}}
[Worker] Message processed for correlation_id=3ede5265-3c97-4b11-b198-2f9991494533: {"success": true, "userId": "30a9d728-a2cd-4ce4-ae98-d22037615796", "message": "User created"}
[Worker] Response sent to user_responses: {"correlation_id": "3ede5265-3c97-4b11-b198-2f9991494533", "data": {"success": true, "userId": "30a9d728-a2cd-4ce4-ae98-d22037615796", "message": "User created"}}
[Worker] Message received from user_requests: {"correlation_id": "5d70ff2a-d0e5-4248-8bb9-510e7a0dd15b", "reply_to": "user_responses", "data": {"operation": "GET_USER", "userId": "30a9d728-a2cd-4ce4-ae98-d22037615796"}}
[Worker] Message processed for correlation_id=5d70ff2a-d0e5-4248-8bb9-510e7a0dd15b: {"success": true, "user": {"userId": "30a9d728-a2cd-4ce4-ae98-d22037615796", "name": "Alice", "email": "alice@test.com", "age": 25}}
[Worker] Response sent to user_responses: {"correlation_id": "5d70ff2a-d0e5-4248-8bb9-510e7a0dd15b", "data": {"success": true, "user": {"userId": "30a9d728-a2cd-4ce4-ae98-d22037615796", "name": "Alice", "email": "alice@test.com", "age": 25}}}

```

PART-2

exact Kafka container name for inbox, tasks, draft, final :

```

• (ds) (ds_venv) Abhinita Sanabada app % docker exec -it homework-8-kafka-1 kafka-topics --create --topic inbox --partitions 1 --replication-factor 1 --bootstrap-server localhost:9092
Created topic inbox.
• (ds) (ds_venv) Abhinita Sanabada app % docker exec -it homework-8-kafka-1 kafka-topics --create --topic tasks --partitions 1 --replication-factor 1 --bootstrap-server localhost:9092
Created topic tasks.
• (ds) (ds_venv) Abhinita Sanabada app % docker exec -it homework-8-kafka-1 kafka-topics --create --topic drafts --partitions 1 --replication-factor 1 --bootstrap-server localhost:9092
Created topic drafts.
• (ds) (ds_venv) Abhinita Sanabada app % docker exec -it homework-8-kafka-1 kafka-topics --create --topic final --partitions 1 --replication-factor 1 --bootstrap-server localhost:9092
Created topic final.
○ (ds) (ds_venv) Abhinita Sanabada app %

```

```

• (ds) (ds_venv) Abhinita Sanabada app % docker exec -it homework-8-kafka-1 kafka-topics --list --bootstrap-server localhost:9092
_inbox
_drafts
_final
_inbox
_tasks
_user_requests
_user_responses
○ (ds) (ds_venv) Abhinita Sanabada app %

```

send_question.py

```
user_requests
user_responses
• (ds) Abhinata Sanabada Homework-8 % python send_question.py
[Sender] Sent to inbox: {"question_id": "1f126b4d-e6a7-4af2-a52e-0b3fa6570769", "question": "What is Kafka and why is it useful?"}
❖ (ds) Abhinata Sanabada Homework-8 %
```

Planner terminal:

```
• (ds) (ds_venv) Abhinata Sanabada Homework-8 % cd /Users/tachyon_hunter/Desktop/sem-3/DS-Slides/DS/Homework-submission/Homework-8
○ (ds) (ds_venv) Abhinata Sanabada Homework-8 % python planner.py
[Planner] Listening on topic: inbox
[Planner] Received from inbox: {"question_id": "1f126b4d-e6a7-4af2-a52e-0b3fa6570769", "question": "What is Kafka and why is it useful?"}
[Planner] Sent to tasks: {"question_id": "1f126b4d-e6a7-4af2-a52e-0b3fa6570769", "question": "What is Kafka and why is it useful?", "plan": ["Understand the user question", "Identify the key topic and intent", "Prepare a short and clear answer"]}
□
```

Writer terminal:

```
Abhinata Sanabada Homework-8 % conda activate ds
(ds) Abhinata Sanabada Homework-8 % which python
python -m pip -V
/opt/homebrew/Caskroom/miniforge/base/envs/ds/bin/python
pip 26.0.1 from /opt/homebrew/Caskroom/miniforge/base/envs/ds/lib/python3.11/site-packages/pip (python 3.11)
(ds) Abhinata Sanabada Homework-8 % python writer.py
Disabling PyTorch because PyTorch >= 2.4 is required but found 2.2.2
PyTorch was not found. Models won't be available and only tokenizers, configuration and file/data utilities can be used.
[Writer] Listening on topic: tasks
[Writer] Received from tasks: {"question_id": "1f126b4d-e6a7-4af2-a52e-0b3fa6570769", "question": "What is Kafka and why is it useful?", "plan": ["Understand the user question", "Identify the key topic and intent", "Prepare a short and clear answer"]}
[Writer] Sent to drafts: {"question_id": "1f126b4d-e6a7-4af2-a52e-0b3fa6570769", "question": "What is Kafka and why is it useful?", "plan": ["Understand the user question", "Identify the key topic and intent", "Prepare a short and clear answer"], "draft": "Kafka is a distributed event streaming platform used to move messages between systems reliably and at scale. It helps decouple services and supports real-time communication through topics."}
□
```

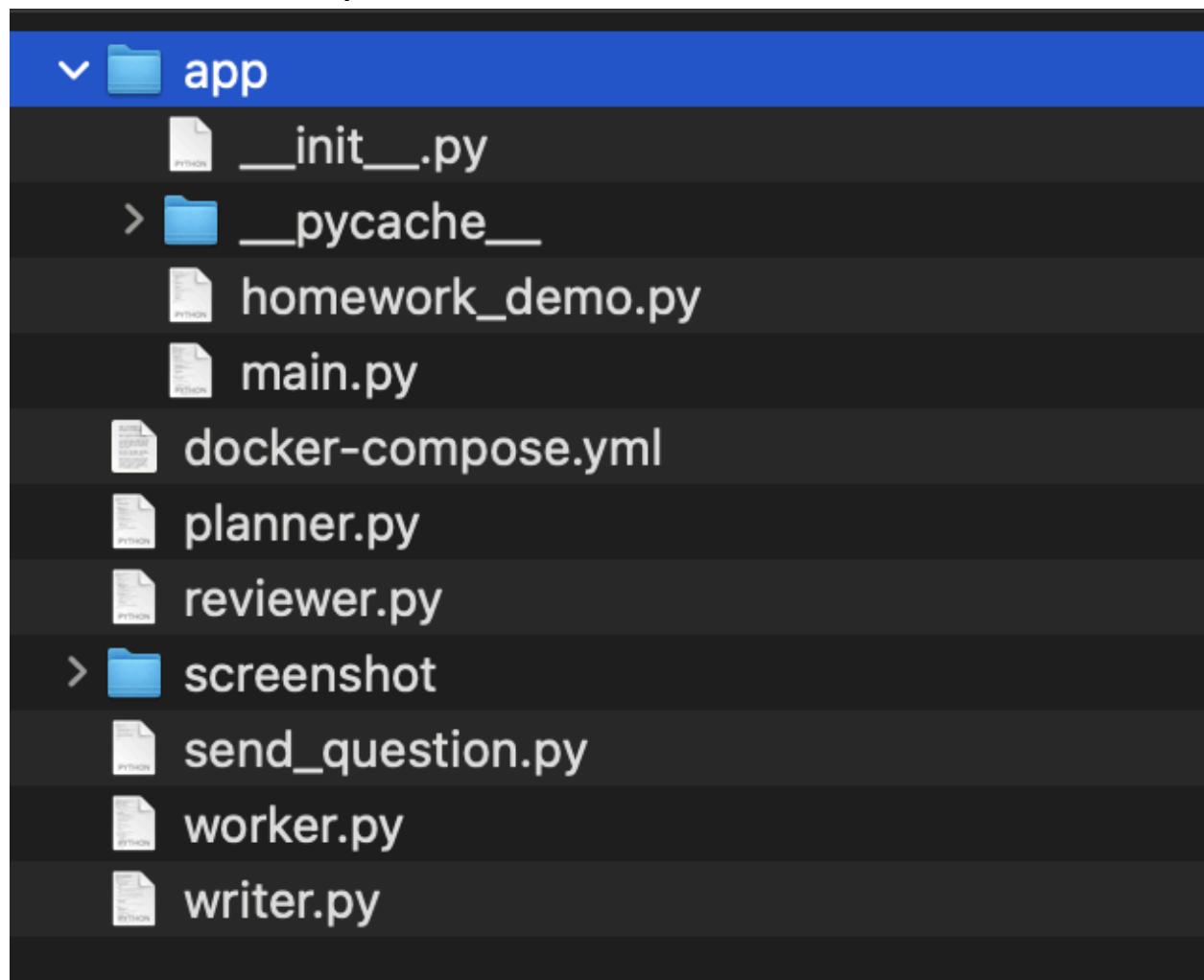
Reviewer terminal:

```
(ds) (ds_venv) Abhinata Sanabada Homework-8 % python reviewer.py
[Reviewer] Listening on topic: drafts
[Reviewer] Received from drafts: {"question_id": "1f126b4d-e6a7-4af2-a52e-0b3fa6570769", "question": "What is Kafka and why is it useful?", "plan": ["Understand the user question", "Identify the key topic and intent", "Prepare a short and clear answer"], "draft": "Kafka is a distributed event streaming platform used to move messages between systems reliably and at scale. It helps decouple services and supports real-time communication through topics."}
[Reviewer] Sent to final: {"question_id": "1f126b4d-e6a7-4af2-a52e-0b3fa6570769", "status": "approved", "answer": "Kafka is a distributed event streaming platform used to move messages between systems reliably and at scale. It helps decouple services and supports real-time communication through topics."}
□
```

Final topic output:

```
work-8/
○ (ds_venv) Rishabhs-MacBook-Air:Homework-8 tachyon_hunter$ docker exec -it homework-8-kafka-1 kafka-console-consumer --bootstrap-server localhost:9092 --topic final --from-beginning
{"question_id": "1f126b4d-e6a7-4af2-a52e-0b3fa6570769", "status": "approved", "answer": "Kafka is a distributed event streaming platform used to move messages between systems reliably and at scale. It helps decouple services and supports real-time communication through topics."}
□
```

Folder structure for both part-1 and 2:



A short report explaining how Kafka helped your agents coordinate:

I used Kafka as the communication layer between the Planner, Writer, and Reviewer agents. Instead of having the agents call each other directly, I made each one send and receive messages only through Kafka topics. The Planner read the question from the `inbox` topic and sent a plan to the `tasks` topic. The Writer then consumed the message from `tasks`, generated a short draft answer using LangChain, and published that draft to `drafts`. Finally, the Reviewer consumed the draft from `drafts`, checked it, and sent the approved result to the `final` topic.

Using Kafka helped me organize the system into separate steps, where each agent had a clear role and only focused on its own task. It also made the workflow easier to track, because I could see the messages moving through each topic and verify that every stage was working correctly. Overall, Kafka helped coordinate the agents in a clean and reliable way by passing information from one stage to the next without direct dependency between the agents.